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Non-linear model-based estimation of quadratic and cubic damping mechanisms governing the dynamics of a chaotic spherical pendulum

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Abstract

We investigate the non-linear damping mechanisms that govern the dynamics of a chaotic spherical pendulum. We reproduce the conditions of the celebrated chaotic experiment of Tritton (1986), and identify the bifurcation regions that correspond to periodic, quasi-periodic and chaotic-like response. Construction of the pendulum frequency and damping backbone curves from free vibration decay data reveal the existence of non-negligible non-linear damping. A non-linear model-based estimation procedure enables extraction of both quadratic and cubic damping coefficients which are deduced from a non-linear Rayleigh-type gradient dissipation function. Comparison of experimental results with solutions obtained via numerical analysis of the strongly non-linear pendulum model, augmented by the estimated damping mechanisms, sheds light on the possible cause of documented discrepancies in similar whirling systems between measurements and theoretical predictions obtained from models with only linear damping.

Keywords

Bifurcation structure, chaotic dynamics, model-based estimation, non-linear damping, spherical pendulum

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1. Introduction

The periodically excited spherical pendulum is one of the fundamental non-linear dynamical systems that exhibit a transition from simple periodic motion in the plane of excitation, to a complex spatio-temporal bifurcation structure that includes coexisting periodic and quasi-periodic motion, culminating with chaotic dynamics that are sensitive to initial conditions. The first to theoretically determine this complex and chaotic bifurcation structure near its primary resonance was Miles (1984) who asymptotically and numerically analyzed the averaged equations of a linearly damped spherical pendulum truncated to cubic order. Soon after, Tritton (1986) qualitatively validated the findings of Miles experimentally, and postulated that the discrepancy between theory and experiment was due to the presence of large air resistance of his 62 mm diameter sphere that cannot be accounted for by a linear damping model. Since the celebrated Tritton experiment, there has been additional documentation of laboratory experiments (Irons, 1990; Kana and Fox, 1995; Todd and Vohra, 1998) that revealed an identical chaotic bifurcation structure for the spherical pendulum.

However, the documented theoretical analysis of the corroborating chaotic experiments has all been based on geometrically non-linear Hamiltonian models augmented by linear damping (Bryant, 1993; Kana and

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Fox, 1995; Todd and Vohra, 1998; Tritton and Groves, 1999; Cartwright and Tritton, 2010).

The documentation of non-linear damping in dynamical systems undergoing flow-structure interaction is not new. Examples include viscous effects that yield for high Reynolds numbers a resistance proportional to the square of the velocity (Tritton, 1977) that governs aeroelastic drag (Dowell and Tang, 2003) and ship roll damping (Kleiman and Gottlieb, 2009). In addition to the common two-term linear and quadratic formulations first proposed by Froude (1955), equivalent linear and cubic damping models have also been analyzed asymptotically (Cardo et al., 1982) and globally (Bikdash et al., 1994).

Thus, in order to bridge the gap between the documented discrepancies between experimental measurements and corresponding theoretical analysis of the chaotic spherical pendulum, it is the objective of this research to propose a consistent damping model that includes linear, quadratic and cubic damping mechanisms and to estimate their properties from a set of free vibration decay and periodically forced vibration experiments. There exist an abundance of procedures that enable parametric identification of linear and non-linear dynamical systems that are excited externally or parametrically (Liang and Feeny, 2006a and references within) and estimation of damping parameters via energy balancing (Liang and Feeny, 2006b). We note that non-linear parameter estimation from free vibration decay has been successfully done via an energy rate balance approach (Roberts, 1985), a Hilbert transform based methodology (Feldman, 1994, 2007) and recently a spline-based slowly varying envelope analysis that enables direct construction of the system frequency and damping backbone curves (Gottlieb et al., 2007; Senesh et al., 2010).

The paper is organized as follows. We describe the experimental analysis in section 2 where we reconstruct the Tritton experiment, and identify the bifurcation regimes that correspond to periodic, quasi-periodic and chaotic-like response. Construction of the pendulum frequency and damping backbone curves from free vibration decay data reveals the existence of non-negligible non-linear damping. We derive a theoretical geometrically non-linear model in section 3 which consistently incorporates a Rayleigh-type gradient dissipation function that includes linear, quadratic and cubic components. We propose an asymptotic model-based estimation procedure to determine the non-linear damping parameters in section 4, and validate the procedure by comparison of experimental results with solutions obtained via numerical analysis of the strongly non-linear pendulum model that incorporate the optimized damping parameters in section 5. We close in the final section with a short discussion of our results and their implications to similar whirling spatio-temporal systems.

2. Experimental analysis

2.1. The experimental set-up

The experiments were performed in the Technion Center for Material Mechanics. The experimental set-up (Figure 1) included a shaker, a pendulum, a digital video and computer for data acquisition (Habib, 2008). The shaker linearly moved the top of the pendulum periodically with a fixed amplitude of $a = 7.68$ mm in a frequency range between 0.8 and 2 Hz. The amplitude and frequency of the excitation were measured directly by a linear variable differential transformer (LVDT) mounted on the shaker. The experimental data was integrated in a standard LabView processor (2000).

Following the Tritton experiment (1986), the pendulum components included a thin nylon wire with a diameter of 0.4 mm and length of 276 mm connected to a smoothed and blackened tennis ball with a diameter of 60 mm. The data acquisition was obtained by a standard digital video with an image frequency resolution of 30 frames per second.

The experimental procedure included digital video recordings of the pendulum in free vibration decay and forced vibration motion. In free vibration the camera offered a perpendicular side-view of the pendulum, whereas for forced vibration, a ladder mounted camera offered a top-view which enabled pendulum tracking for both planar and out-of-plane motion. In order to get numerical data from the videos, we defragged the videos in frames, and retrieved the sphere planar coordinates in each frame. Conversion of raw data (pixel) to actual displacement was done by automatic calibration of the known sphere diameter.

We note that a systematic error was introduced by the top-view recording of sphere motion which traversed different vertical planes (e.g. the larger oscillation amplitude resulted in a larger sphere in the respective video frame). Consequently, in order to minimize distortion, in each video the sphere dimensions were calibrated when it was closest to the camera position. The free decay video recordings were not influenced by this error because the camera offered a side view of the pendulum, leaving the ball always in the same plane.

2.2. Forced vibration bifurcation structure

The fundamental bifurcation structure of the periodically excited Tritton experiment was readily reproduced (Figure 2) and includes unique planar periodic motion ($f < 0.865$ Hz and $f > 0.95$ Hz), coexisting planar and spatial periodic motion ($0.94 < f < 0.95$ Hz), what appeared to be unique spatial periodic motion ($0.92 < f < 0.94$ Hz), coexisting spatial periodic and

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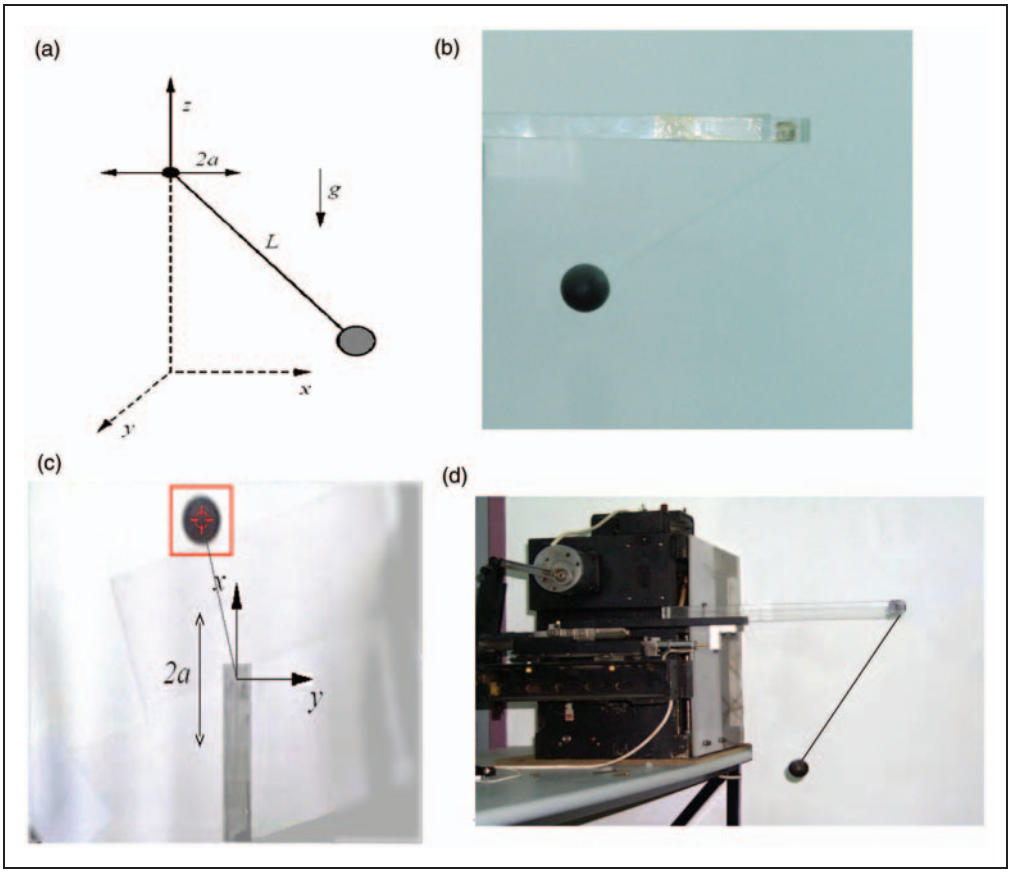


Figure 1. (a) Definition sketch of spherical pendulum system, (b) the physical pendulum, (c) example video frame for coordinate capturing (top view), (d) experimental set-up (shaker, LVDT, horizontal bar, string, pendulum).

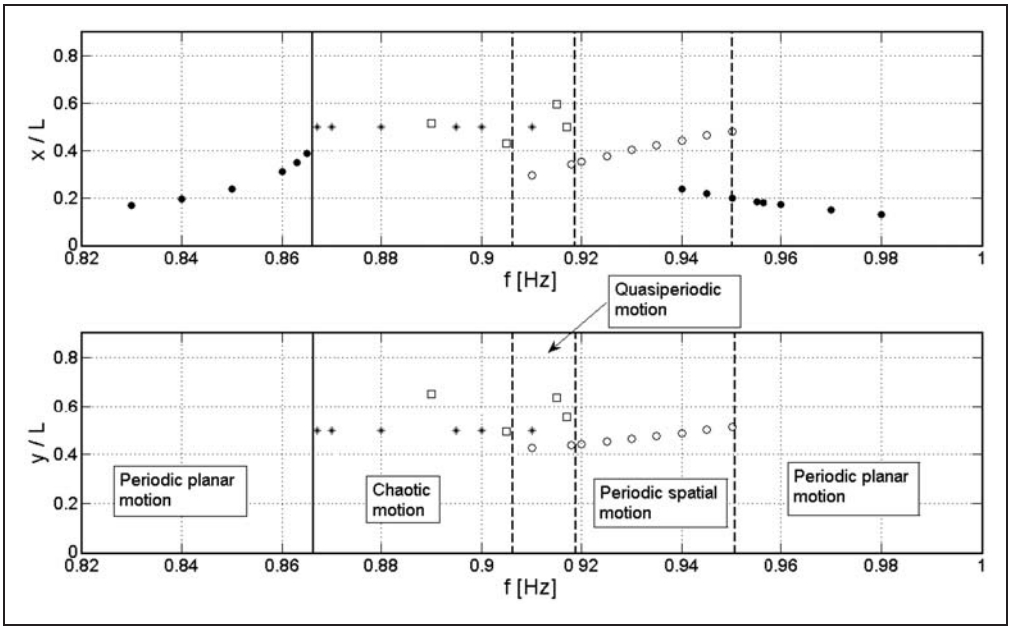


Figure 2. Comparison of the experimental and numerical bifurcation structures depicting in-plane (top) and out-of-plane (bottom) response. Vertical lines correspond to experimental thresholds separating the different types of motion. The numerical results include periodic planar motion (dark circles), periodic spatial motion (open circles) quasi-periodic motion (stars) and chaotic-like motion (stars).

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quasi-periodic motion ($0.91 < f < 0.92$ Hz), a range of mixed quasi-periodic and chaotic-like motion ($0.89 < f < 0.91$ Hz) and a range of what appeared to be unique chaotic-like motion ($0.865 < f < 0.89$ Hz). We note that the seven region bifurcation structure in Figure 2 is similar to the eight region structure previously obtained asymptotically and experimentally (Miles, 1984; Tritton, 1986) but differs from the eleven region structure later obtained by numerical integration of the Miles averaged equations (Tritton and Groves, 1999: Table 1). The intricate bifurcation structure observed by Tritton and Groves included both quasi-periodic and chaotic solution reversals that relate to changes between clockwise and anticlockwise whirling directions which are similar to the transit and non-transit motions of the whirling string (O'Reilly and Holmes, 1992; Rubin and Gottlieb, 1996).

Examples of typical planar periodic, spatial periodic, quasi-periodic and chaotic-like solutions are depicted by their normalized measured time-series x(t), y(t) and physical phase plane projections y(x) in the left-hand side of Figures 3, 4, 5, and 6, respectively. We note the experimental results were recorded after transient decay (denoted $t = 0$) up to a maximal time

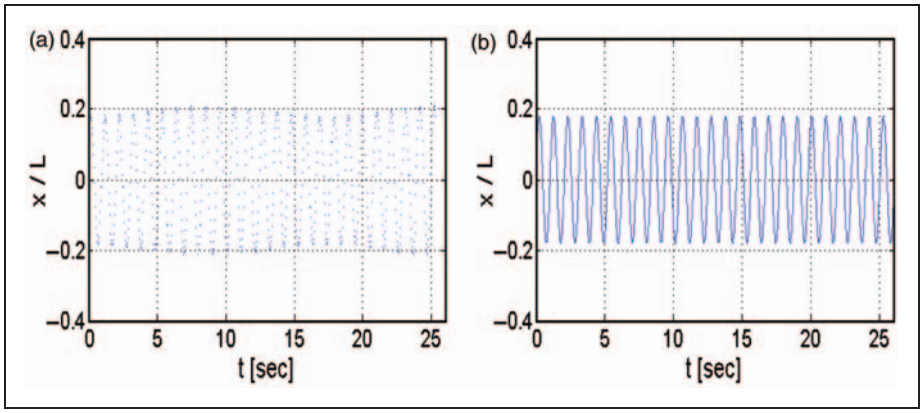


Figure 3. Comparison of experimental (left) and numerical (right) time series for periodic planar motion ($y/L = 0$) measured at $f = 0.956$ Hz ($\Omega = 1.0622$).

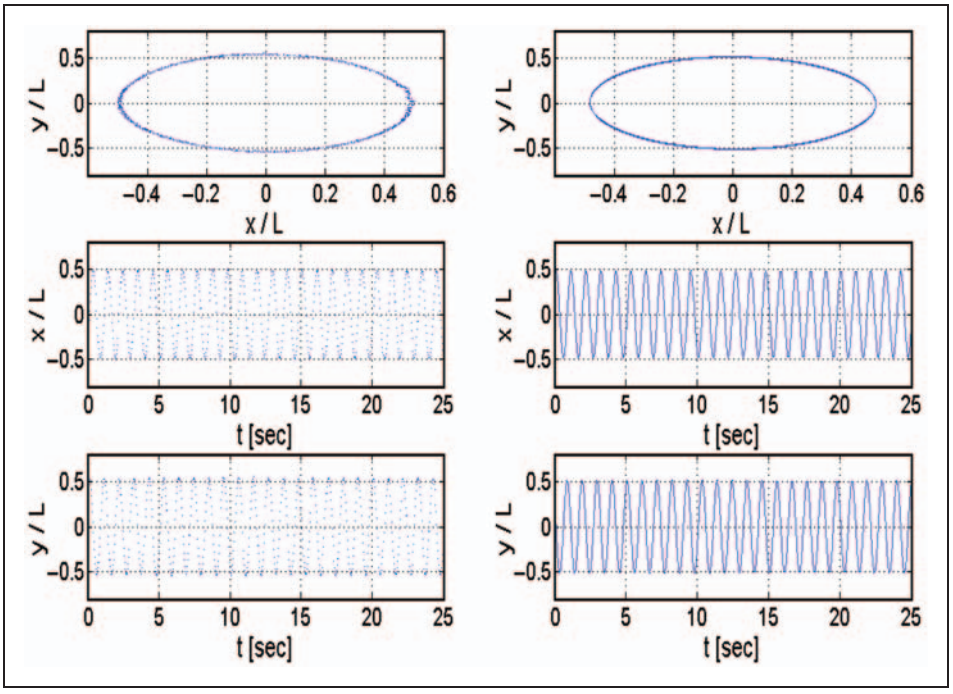


Figure 4. Comparison of experimental (left) and numerical (right) phase-plane (top) and time series (middle and bottom) for periodic spatial motion measured at $f = 0.95$ Hz ($\Omega = 1.0555$).

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corresponding twenty or fifty input periods for the periodic (Figures 3 and 4) and a-periodic (Figures 5 and 6) motions, respectively. Furthermore, for convenience of comparison, the right-hand side of Figures 3 to 6 depicts the corresponding figures obtained from numerical integration of the theoretical model derived in section 3 and discussed in section 5.

2.3. Free vibration decay

After verification of the fundamental chaotic pendulum bifurcation structure, we obtained the pendulum planar free vibration decay by overlapping of several different experiments. The overlapping of experiments enabled generation of a longer time-series than that limited by the digital camera memory (5000 frames or 166 seconds). The instantaneous amplitude envelope defined by localized maxima and minima is depicted in Figure 7(a). We then computed the instantaneous frequency (Figure 7(b)) by the harmonic mean of the three opposite consecutive instantaneous periods (Gottlieb et al., 2007; Senesh et al., 2010):

$$F = \left(T_{i-1}^{-1} + T_i^{-1} + T_{i+1}^{-1}\right)/3 \tag{1}$$

We note that Figure 7(b) clearly depicts the range of linear free vibration ($100 < t < 300$ sec) where

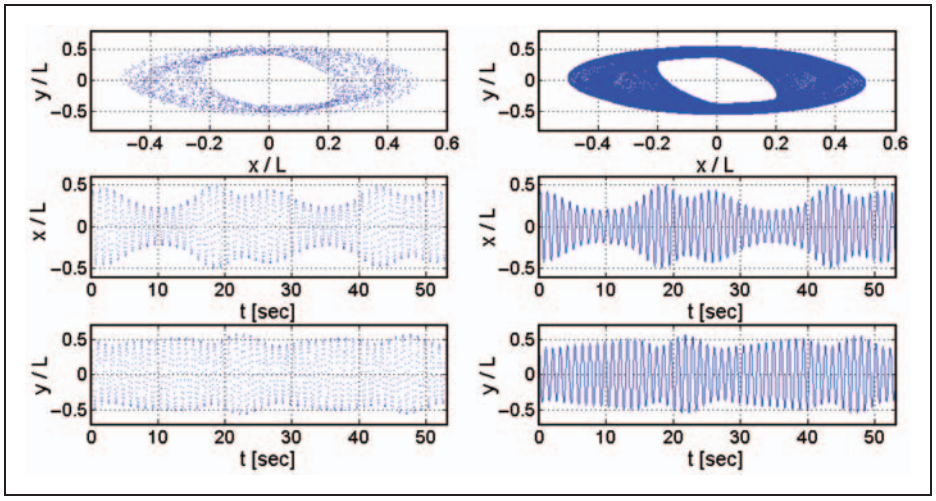


Figure 5. Comparison of experimental (left) and numerical (right) phase-plane (top) and time series (middle and bottom) for quasi-periodic spatial motion measured at f = 0.917 Hz ($\Omega = 1.0189$).

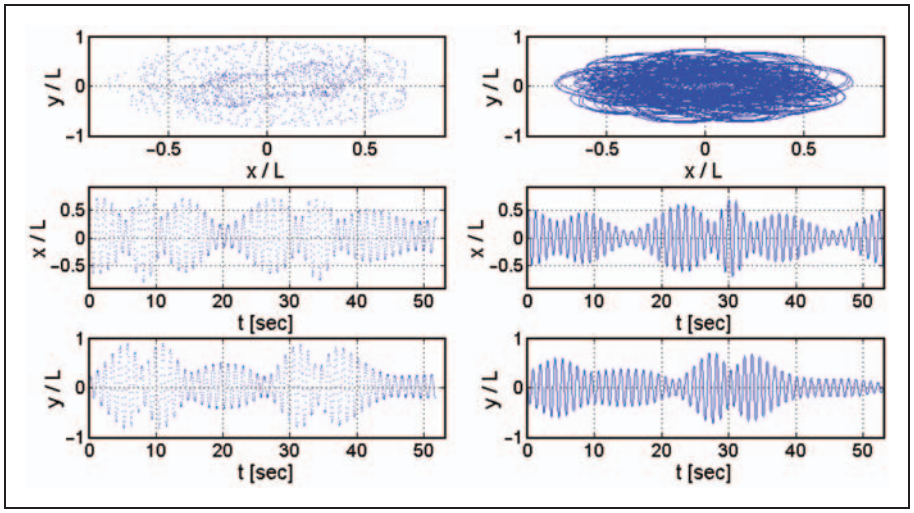


Figure 6. Comparison of experimental (left) and numerical (right) phase-plane (top) and time series (middle and bottom) for chaotic-like spatial motion measured at f = 0.90 Hz ($\Omega = 1.0$).

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on average, the instantaneous frequency (F) is constant and yields the system damped natural frequency ($f_d = 0.9$ Hz). We note that the wide spread of points in this range ($0.1 < |x/L| < 0.3$) is primarily due to the sampling rate (30 Hz) which gave an average resolution of 33 points per period of oscillation and thus controlled accuracy of identification of local maxima.

The system frequency backbone curve (Nayfeh and Mook, 1979) depicted in Figure 7(c) is obtained by plotting the instantaneous maximal and minimal amplitudes (A) versus their corresponding frequencies (F). We note that as expected, the frequency backbone exhibits the standard softening behavior of a planar pendulum which in the range of finite displacements will motivate use of a weakly non-linear model in section 4.

The damping backbone curve, first proposed by Roberts (1985), made use of cubic-spline interpolation to determine a relationship between a damping function and measured ship roll amplitude decay. Gottlieb and Feldman (1997) later employed a Hilbert transform to obtain a similar functional relationship between the decay amplitude (A) and its corresponding instantaneous damping ratio ($\tilde{\zeta}$). We employ here the latter damping backbone definition, which for convenience is derived by a harmonic balance procedure in Appendix A:

$$\tilde{\zeta} = -[A_t/A + F_t/(2F)]/(2\pi F) \qquad (2)$$

where the localized synchronous derivatives of the slowly varying amplitude (A_t) and frequency (F_t) are obtained numerically (via a finite difference procedure) from spline functions of the identified maximal and minimal amplitudes (A) and corresponding instantaneous frequencies (F). The damping backbone in Figure 7(d) reveals a non-negligible non-linear content that is reminiscent of the backbones obtained from ship roll models with linear-plus-cubic damping moment (Roberts, 1985) or a linear-plus-quadratic damping (Gottlieb and Feldman, 1997). We note that from Figure 7(d) we can evaluate the linear damping content as the zero intersect ($A = 0$) of the mean curve though the data (e.g. $\zeta_n = \tilde{\zeta}(A = 0) < 0.001$). An alternative method to validate the bounds on the pendulum linear damping ratio is via a logarithmic decrement (Meirovitch, 1986) which is defined by the ratio between two consecutive displacements separated by a damped period ($d = \ln(x_1/x_{j+1})/j$). Substitution of the computed value from the data in Figure 7(a) ($d = 0.0065$) into the relation between decrement and damping ratio ($\zeta_n = d[(2\pi)^2 + d^2]^{-1/2}$) yields $\zeta_n = 0.0011$ which is consistent with the backbone estimate of linear damping content in our experiment. We note that due to the large spread of data in the small amplitude range, we can only treat this damping ratio as an upper bound.

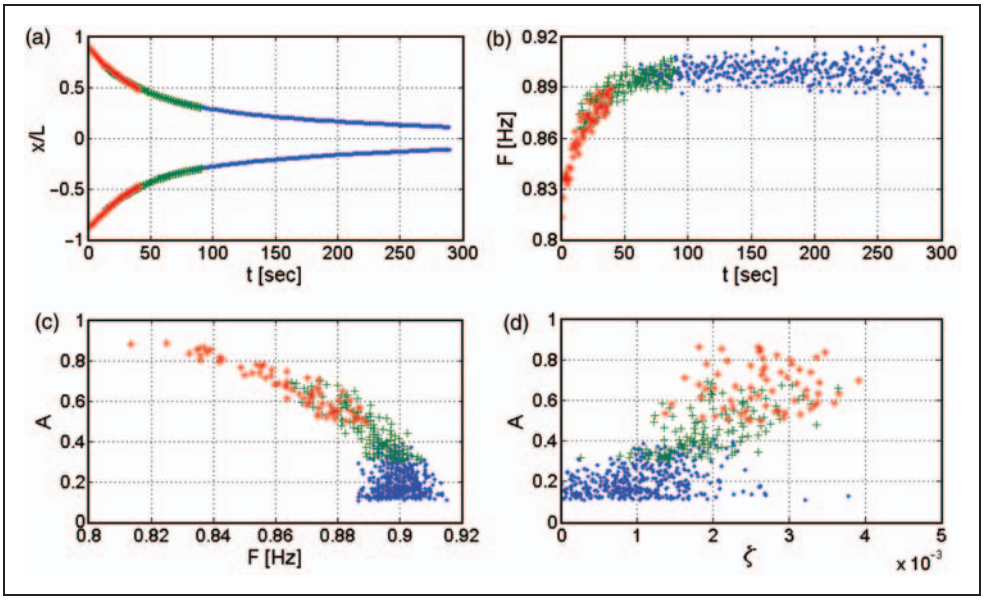


Figure 7. Analysis of free vibration decay data: (a) amplitude envelope, (b) frequency envelope, (c) frequency backbone curve, (d) damping backbone curve.

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3. Theoretical analysis

3.1. Non-linear model derivation

We consider the spherical pendulum subject to harmonic base excitation in the horizontal direction (Figure 1) and employ a Lagrangian formulation (Meirovitch, 1988) where the system kinetic and potential energies in a Cartesian frame of reference $[KE = m(x_t^2 + y_t^2 + z_t^2)/2, \quad PE = mgz]$ can readily be reduced via the holonomic length constraint $[z = L - (L^2 - (x^2 + y^2))^{1/2}]$, to the following form with two generalized coordinates $q_j = (x,\ y)$:

$$KE = \frac{m}{2}\left(x_t^2 + y_t^2 + \frac{(RR_t)^2}{L^2 - R^2}\right),$$
$$PE = mg \cdot \left(L - \sqrt{L^2 - R^2}\right), \tag{3}$$

where the m, L, g are the pendulum mass, length and acceleration of gravity, respectively; the subscript t denotes differentiation with respect to time, and the auxiliary functions $(R^2,\ RR_t)$ are:

$$R^2 = \left((x-u)^2 + y^2\right), \quad RR_t = ((x-u)(x_t - u_t) + yy_t). \tag{4}$$

We consider a harmonic function of time for horizontal base excitation:

$$u = a\cos(\omega_a t) \tag{5}$$

where a and ω_a are the amplitude and frequency of excitation, respectively.

The generalized damping force is derived from a three-term gradient Rayleigh dissipation function $(Q_j = -(\partial G/\partial q_{jt})$where j = x, y) which incorporates the relative velocity in the horizontal direction (Gottlieb, 1997) and ensures that the force will always remain dissipative:

$$G = D^2\left(\frac{b}{2} + \frac{c}{3}D + \frac{d}{4}D^2\right),$$
$$D^2 = \left((x_t - u_t)^2 + y_t^2 + \frac{(RR_t)^2}{L^2 - R^2}\right) \tag{6}$$

We rescale the generalized coordinates by the pendulum length (L) and rescale time $(\tau = \omega_n t)$ by its linear natural frequency $(\omega_n = \sqrt{g/L})$. Substitution of equations (3) to (6) with the corresponding length and time scales into a Lagrangian formulation $[(\partial KE/\partial q_{j\tau}) - (\partial KE/\partial q_j) + (\partial PE/\partial q_j) = -(\partial G/\partial q_{j\tau})]$, and decoupling of the inertial terms (due to $z_\tau = z_\tau(x,\ y,\ x_\tau,\ y_\tau)$in the kinetic energy), yields the following equations of motion for the spherical pendulum:

$$x_{\tau\tau} + E(x-u) = -F(x_\tau - u_\tau)$$
$$y_{\tau\tau} + Ey = -Fy_\tau \tag{7}$$

where the subscript defines derivative with respect to non-dimensional time τ, and the conservative (E) and damping (F) terms are:

$$E = \left[\sqrt{1-R^2} - (x-u)u_{\tau\tau} + (x_\tau - u_\tau)^2 + y_\tau^2 + (RR_\tau)^2/(1-R^2)\right]$$
$$F = \left[\beta + \gamma\left((x_\tau - u_\tau)^2 + y_\tau^2\right)^{1/2} + \delta\left((x_\tau - u_\tau)^2 + y_\tau^2\right)\right]. \tag{8}$$

The conservative term E (deduced from the derivatives of the kinetic and potential energies) includes the following non-dimensional auxiliary terms $(R^2,\ RR_\tau)$:

$$R^2 = \left((x-u)^2 + y^2\right), \quad RR_\tau = ((x-u)(x_\tau - u_\tau) + yy_\tau). \tag{9}$$

We note that we have ignored the small contribution of the auxiliary term in the dissipation function $[z_\tau^2 = (RR_\tau)^2/(1-R^2)]$, and that the scaled base excitation is:

$$u = \alpha\sin(\Omega\tau) \tag{10}$$

where the non-dimensional damping parameters are $\beta = (b/m\omega_n)$, $\gamma = (cL/m)$, $\delta = (dL^2\omega_n/m)$, and the non-dimensional amplitude and frequency are $\alpha = (a/L)$ and $\Omega = (\omega_a/\omega_n)$, respectively. Thus, equations (7) to (10) define the strongly non-linear spherical pendulum dynamical system which is governed by five non-dimensional parameters: amplitude and frequency of excitation $(\alpha,\ \Omega)$ and linear, quadratic and cubic damping coefficients $(\beta,\ \gamma,\ \delta)$. In our analysis, we fix the amplitude parameter according to the fixed value of the experiment (7.68 mm) to yield $\alpha = 0.0251$. We note that this parameter is larger than the corresponding value in the Tritton experiment $(\alpha = 0.00965)$. Furthermore, linearization of the system about its stable equilibrium (x = y = 0) yields a non-dimensional natural frequency of unity and a corresponding linear damping ratio of $\zeta_n = \beta/2$.

3.2. Forced vibration

Numerical integration of the equations of motion with a selected set of damping parameters reveals a seven region bifurcation structure which includes stability thresholds that are similar to the experimental ones in Figure 2. Examples of spatial periodic, quasi-periodic

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and chaotic solutions that are obtained numerically are depicted in Figures 8, 9, and 10, respectively. These figures include the pendulum time-series x(t), its power spectra S(f), the physical phase-plane projection y(x), and its corresponding Poincaré map (Y(X)) which is defined by stroboscopically sampling the time-series each forcing period (Guckenheimer and Holmes, 1983; Nayfeh and Balachandran, 1995). These examples are obtained by numerical integration of the strongly non-linear system with the experimental value for base excitation ($\alpha = 0.0251$) and a selected set of damping coefficients ($\beta = 0.0009$, $\gamma = 0.006$, $\delta = 0.0045$) that will be shown in section 4 to best fit the experimental conditions. We note that the power spectrum of the periodic solution incorporates distinct odd harmonics (Figure 8), and that both spectra of the quasi-periodic and chaotic solutions (Figures 9 and 10) are wide-banded. Furthermore, the quasi-periodic Poincaré map forms a closed topological curve in the physical phase-plane projection (Figure 9) whereas its chaotic counterpart (Figure 10) reveals an irregular distribution of points.

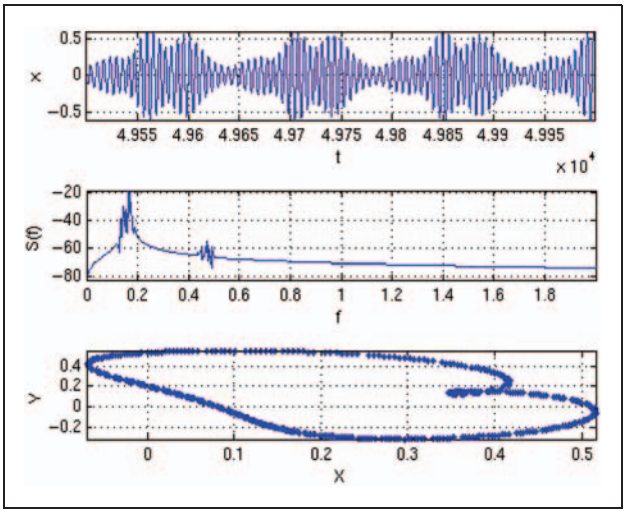


Figure 9. Spatial quasi-periodic solution at $\Omega = 1.0189$: time-series (top), power spectra (middle), Poincaré map projection (bottom).

4. Model-based parameter estimation

In order to estimate the non-linear damping parameters from free vibration decay, we expand the equation of motion for the planar pendulum ($y = 0$) to cubic order for the unforced case ($\alpha = 0$):

$$x_{\tau\tau} + \left(\beta + \gamma|x_\tau| + \delta x_\tau^2\right)x_\tau + \left(1 + x_\tau^2 - x^2/2\right)x = 0 \quad (11)$$

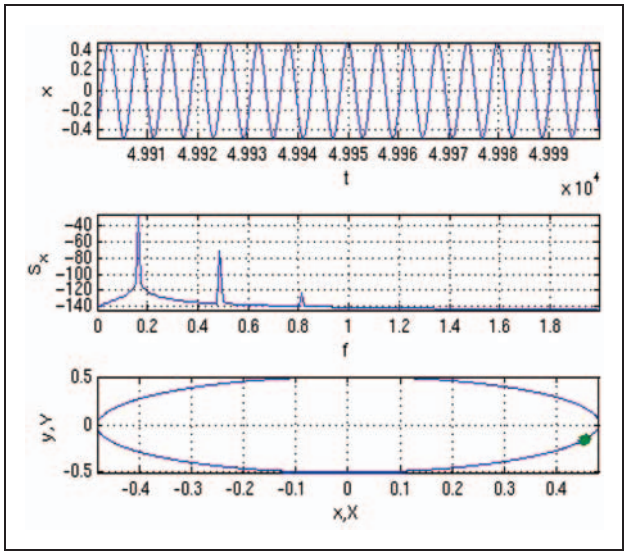


Figure 8. Spatial periodic solution at $\Omega = 1.055$: time-series (top), power spectra (middle), phase-plane projection and Poincaré map (bottom).

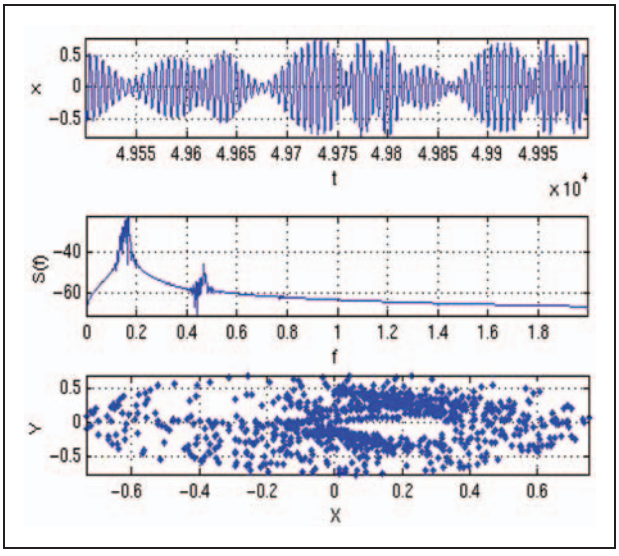


Figure 10. Spatial chaotic solution at $\Omega = 1.0$: time-series (top), power spectra (middle), Poincaré map projection (bottom).

For the case of small displacements ($x = \varepsilon q$, where $\varepsilon \ll 1$ is a small parameter), the free vibration decay equation (11) can readily be rewritten in the form that is ready for averaging:

$$q_{\tau\tau} + q = -\varepsilon^2\left[\left(1 + q_\tau^2 - q^2/2\right)q + \left(\hat{\beta} + \hat{\gamma}|q_\tau| + \hat{\delta}q_\tau^2\right)q_\tau\right] \quad (12)$$

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where we balance the damping parameters so that the right-hand-side of equation (12) is of the same order ($\beta = \varepsilon^2\hat{\beta}$, $\gamma = \varepsilon\hat{\gamma}$, $\delta = \hat{\delta}$). In anticipation of a non-linear decaying solution, we assume a solution form of $x = \varepsilon a\cos(\omega\tau + \varphi)$ [or $q = a\cos(\omega\tau + \varphi)$], where the amplitude and phase are slowly varying [$a = a(\varepsilon\tau)$, $\phi = \phi(\varepsilon\tau)$], and the oscillation frequency is a small perturbation of the undamped natural frequency [i.e. $(1-\omega^2) \sim O(\varepsilon^2)$]. Consequently, application of an averaging procedure (Sanders and Verhulst, 1985), results in the following evolution equations for the slowly varying amplitude and phase:

$$a_\tau = -\left[\left(\frac{\beta}{2}\right) + \left(\frac{4\gamma\omega}{3\pi}\right)a + \left(\frac{3\delta\omega^2}{8}\right)a^2\right]a = -[\tilde{\zeta}]a$$
$$\phi_\tau = \left[\left(\frac{1-\omega^2}{2\omega}\right) - \left(\frac{3-2\omega^2}{16\omega}\right)a^2\right] \tag{13}$$

We note that for negligible detuning (or $\omega \sim 1$), the evolution equations reduce to the limiting case of:

$$a_\tau = -\left[\left(\frac{\beta}{2}\right) + \left(\frac{4\gamma}{3\pi}\right)a + \left(\frac{3\delta}{8}\right)a^2\right]a$$
$$\phi_\tau = -\left(\frac{a}{4}\right)^2 \tag{14}$$

We note that slowly varying phase equation for very small amplitudes yields the linear phase evolution $\phi = (Ct + \phi_0)$ where $C = -(a_0/4)^2$ and a_0, ϕ_0 are initial conditions. Thus, as the solution form is $x = a\cos(\tilde{\omega}\tau + \phi_0)$, the modified frequency is $\tilde{\omega} = (1-(a/4)^2)$. This result is identical to the Hamiltonian approximation for the pendulum natural frequency obtained via the Lindstedt-Poincaré method (Nayfeh and Mook, 1979, p.67). Furthermore, we note that the slowly varying amplitude equation includes combined linear, quadratic and cubic terms that are identical to those reported in literature (Cardo et al., 1982, Nayfeh and Mook, 1979 p.123).

As we are interested in the balance of quadratic and cubic damping coefficients, we construct the damping backbone curve (Gottlieb and Feldman, 1997; Roberts, 1985) by matching the terms on the right-hand-side of equation 13 to the equivalent linear damping relationship. This yields the following implicit form for the damping backbone where we have replaced the amplitude (a) with the measured maximal non-dimensional amplitude A (or maximum |x/L|) and defined the frequency with its non-dimensional value (i.e. $\omega = (f/f_n)$ where $f_n = 0.9$ Hz):

$$\tilde{\zeta} = \left[\left(\frac{\beta}{2}\right) + \left(\frac{4\gamma\omega}{3\pi}\right)A + \left(\frac{3\delta\omega^2}{8}\right)A^2\right] \tag{15}$$

A candidate solution for linear damping is twice the value of the estimated zero intersect in Figure 7(d) or $0.0008 < \beta < 0.001$. Thus, by substitution of several measured values of $(A(\tilde{\zeta}))$ in Figure 7(d) with their corresponding measured frequencies (A, F) from Figure 7(c), we obtain a candidate set of parameters for quadratic and cubic damping. Figure 11 depicts the theoretical damping backbone obtained from equation (15) (dotted line) overlaid on the experimentally computed backbone (red, green and blue points). We improve the tuning of the quadratic and cubic damping parameters by numerically integrating the strongly non-linear system and minimizing the difference between the maximal amplitudes in Figure 12(a). This yields the following set of optimal damping parameters: $\beta = 0.0009$, $\gamma = 0.006$, $\delta = 0.0045$. A comparison of the numerically obtained damping backbone curve (solid line) with the approximation from equation (15) (dotted line) in Figure 11 shows that they are identical for small and finite amplitudes ($0.1 < A < 0.45$) but deviate (up to 12%) for the large amplitudes ($\tilde{\zeta} = 0.003$). Furthermore, a comparison of the numerically obtained free vibration decay envelopes and backbone curves with the computed values of the experiment is shown in Figure 12. We note that while there is an excellent agreement between the numerical and experimental amplitude envelopes (within 1%), the correspondence of the experimental and numerical frequency envelopes in Figure 12(b) is less accurate (within 2.2%) in the small amplitude range.

Consequently, as the experimentally computed damping backbone points in Figure 12(d) require

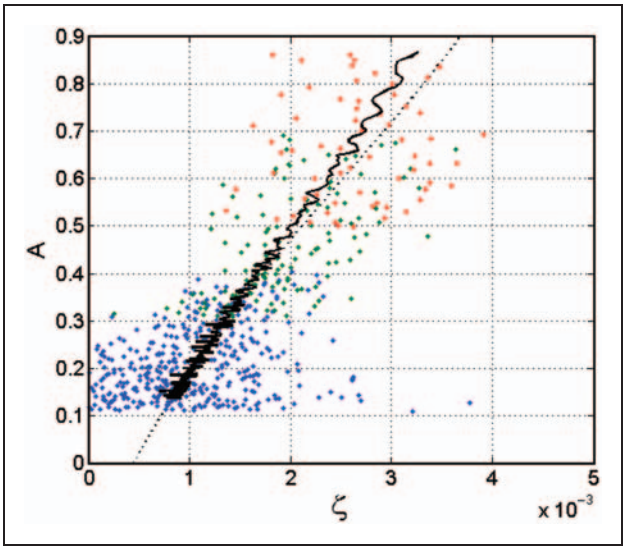


Figure 11. The theoretical damping backbone curve (dotted line) overlaid on the experimentally computed backbone estimate (blue, green, red points)and on the numerically obtained backbone estimate (solid line).

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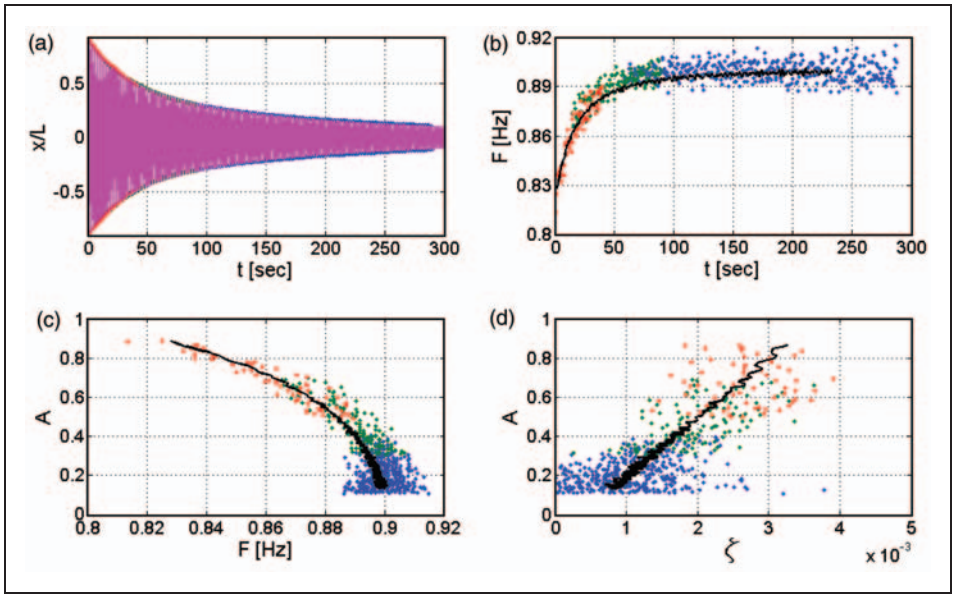


Figure 12. Comparison of numerically obtained and experimentally computed envelopes and backbone curves: (a) amplitude envelope, (b) frequency envelope, (c) frequency backbone curve, (d) damping backbone curve.

approximate amplitude and frequency derivative estimates (e.g. A_t, F_t in equation (2)), comparison of the root-mean-square of the experimental damping backbone points with their numerical counterparts are less accurate (within 3.5%).

5. Validation and prediction of quasi-periodic and chaotic response

5.1. Periodic planar and spatial motion

We further validate the optimal parameter set ($\beta = 0.0009$, $\gamma = 0.006$, $\delta = 0.0045$) via numerical integration of the strongly non-linear equations of motion forced vibration ($\alpha = 0.0251$). A comparison of planar ($\Omega = 1.0622$) and spatial ($\Omega = 1.0555$) periodic experimental and numerical solutions are depicted in Figures 3 and 4, respectively. We note the excellent agreement between experiment and theory for both planar (2%) and spatial (1%) periodic results, respectively.

5.2. Quasi-periodic and chaotic motion

Using the optimal values for the damping coefficients, we numerically integrate the strongly non-linear equations of motion for the conditions of quasi-periodic and chaotic-like response. Figures 5 and 6 depict the comparison between experiment and theory for example quasi-periodic ($\Omega = 1.0189$) and chaotic-like ($\Omega = 1$) solutions. We note that the agreement is excellent for the quasi-periodic solutions (within 1.5%), whereas the maximal magnitude of the irregular chaotic solutions is found to differ slightly (3%) in the direction perpendicular to that of the excitation.

We summarize and note that the stability thresholds in the experimental bifurcation diagram in Figure 2 slightly differ from those obtained by the theoretical model. While we have numerically validated the existence of the seven primary bifurcation regions described in section 2, we note that performing a comprehensive analysis for coexisting solutions and domains of attraction experimentally is extremely difficult as the single control parameter is the excitation frequency which enables only a sweep-up and sweep-down to construct the experimental bifurcation diagram.

6. Closing remarks

In this research we investigated the non-linear damping mechanisms that govern the dynamics of a chaotic spherical pendulum. We reproduced the conditions of the celebrated chaotic experiment of Tritton (1986), and identified the bifurcation regions that correspond to periodic, quasi-periodic and chaotic-like response due to periodic horizontal base excitation, where the amplitude was fixed and the frequency varied close to the pendulum primary resonance. Construction of the pendulum frequency and damping backbone curves from free vibration decay data revealed the existence of non-negligible non-linear damping. We thus derived a Lagrangian based model for the horizontally excited spherical pendulum, augmented by a generalized force deduced from a non-linear Rayleigh-type gradient dissipation function. The resulting damping force included

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linear, quadratic and cubic relative-motion damping mechanisms. A non-linear model-based estimation procedure enabled identification of linear damping and extraction of both quadratic and cubic damping coefficients from the planar system backbone curves. Comparison of experimental solutions with those obtained via numerical analysis of the strongly non-linear pendulum model with the optimized damping parameters, reveals an excellent agreement for both periodic and quasi-periodic solutions, whereas the maximal magnitude of the irregular chaotic solutions was found to differ slightly in the direction perpendicular to that of the excitation. Consequently, we deduce that consistent modelling and validation of non-linear damping mechanisms are essential for prediction of spatio-temporal periodic and quasi-periodic pendulum dynamics and bounds for chaotic response.

This investigation was based on a prescribed excitation amplitude for the spherical pendulum, which revealed a seven region bifurcation structure that includes both periodic, quasi-periodic and chaotic solutions. While, the linear damping parameter is expected to remain within the bounds determined by the free vibration decay analysis, both quadratic and cubic damping parameters are sensitive to the magnitude of the pendulum response which is governed by the amplitude of base excitation. An additional open question is the non-linear damping behavior when there exist bistable periodic solutions, coexisting periodic and chaotic solutions with different maximal amplitudes or intermittent response.

Finally, we recall that additional whirling systems such as taut strings (Johnson and Bajaj, 1989; Leamy and Gottlieb, 2000), suspended cables (Lee and Perkins, 1992; Rega, 2004), rotor dynamics (Inoue and Ishida, 2006; Kligerman et al., 1998) and gyroscopic systems (Perez-Polo and Perez-Molina, 2007) have been shown to exhibit quasi-periodic and chaotic whirling dynamics. Furthermore, we note that several investigations have revealed discrepancies between whirling string experiments and theoretical models that incorporate only linear damping (Nayfeh et al., 1995; O'Reilly and Holmes, 1992; Rubin and Gottlieb, 1996). Thus, we conjecture that additional modelling of multiple non-linear damping mechanisms and accurate model-based estimation of their parameters, may bridge the gap between measurements and theory in both spherical pendulum and similar whirling systems.

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References

Bikdash M, Balachandran B and Nayfeh AH (1994) Melnikov analysis for a ship with a general roll-damping model. *Nonlinear Dynamics* 6: 101–124.

Bryant PJ (1993) Breakdown to chaotic motion of a forced, damped, spherical pendulum. *Physica D* 64: 324–339.

Cardo A, Francescutto A and Nabergoj R (1982) On damping models in free and forced rolling motion. *Ocean Engineering* 9: 295–307.

Cartwright JHE and Tritton DJ (2010) Chaotic dynamics and reversal statistics of the forced spherical pendulum: comparing the Miles equations with experiment. *Dynamicall Systems* 25: 1–16.

Dowell EH and Tang D (2003) *Dynamics of Very High Dimensional Systems*. Singapore: World Scientific.

Feldman M (1994) Nonlinear system vibration analysis using the Hilbert transform: free vibration analysis method. *Mechanical Systems and Signal Processing* 8: 119–227.

Feldman M (2007) Identification of weak nonlinearities in multiple coupled oscillators. *Journal of Sound and Vibration* 303: 357–370.

Froude W (1955) *The Papers of William Froude*. London: The Royal Institution of Naval Architects.

Gottlieb O (1997) Bifurcations of a nonlinear small-body ocean mooring systems excited by finite amplitude waves. *Journal of Offshore Mechanics and Arctic Engineering* 119: 234–238.

Gottlieb O and Feldman M (1997) Application of a Hilbert transform-based algorithm for parameter estimation of a nonlinear ocean system roll model. *Journal of Offshore Mechanics and Arctic Engineering* 119: 239–243.

Gottlieb O, Hoffman A, Wu W, Maimon, R, Edrei R, and Shavit A (2007) The influence of nonlinear air drag on microbeam response for noncontact atomic force microscopy. In: *Proceedings of the 21st ASME Biennial Conference Mechanical Vibration and Noise*, Las Vegas, NV, 4–7 September, DETC–35225, 1–9.

Guckenheimer J and Holmes P (1983) *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*. New York: Springer.

Habib G (2008) Experimental analysis of linear and nonlinear vibrations of a piezoelectric cantilever in several conditions. MSc Thesis, University of Rome-La Sapienza, Rome, Italy.

Inoue T and Ishida Y (2006) Chaotic vibration and internal resonance phenomena in rotor systems. *Journal of Vibration and Acoustics* 128: 156–169.

Irons FE (1990) Concerning the nonlinear behaviour of forced spherical pendulum including the dowsing pendulum. *European Journal of Physics* 11: 107–115.

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Johnson JM and Bajaj AK (1989) Amplitude modulated and chaotic dynamics in resonant string motion. *Journal of Sound and Vibration* 128: 87–107.
Kana DD and Fox DJ (1995) Distinguishing the transition to chaos in a spherical pendulum. *Chaos* 5: 298–310.
Kleiman A and Gottlieb O (2009) Nonlinear dynamics and internal resonances of a ship with a rectangular cross-section in head seas. *Journal of Offshore Mechanics and Arctic Engineering* 131(041301): 1–12.
Kligerman Y, Gottlieb O and Darlow MS (1998) Nonlinear vibration of a rotating system with an electromagnetic damper and a cubic restoring force. *Journal of Vibration and Control* 4: 131–144.
LabView (2000) *Data Acquisition Basics Manual*. Austin, USA: National Instruments.
Leamy MJ and Gottlieb O (2000) Internal resonances in whirling strings involving longitudinal dynamics and material nonlinearities. *Journal of Sound and Vibration* 236: 683–703.
Lee C and Perkins NC (1992) Nonlinear oscillations of suspended cables containing a 2:1 internal resonance. *Nonlinear Dynamics* 3: 465–490.
Liang Y and Feeny BF (2006a) Parametric identification of a base-excited single pendulum. *Nonlinear Dynamics* 46: 17–29.
Liang JW and Feeny FB (2006b) Balancing energy to estimate damping parameters in forced oscillators. *Journal of Sound and Vibration* 295: 988–998.
Meirovitch L (1986) *Elements of Vibration Analysis*. New York: McGraw-Hill.
Meirovitch L (1988) *Methods of Analytical Dynamics*. New York: McGraw-Hill.
Miles JW (1984) Resonant motion of a spherical pendulum. *Physica D* 11: 309–323.
Nayfeh AH and Balachandran B (1995) *Applied Nonlinear Dynamics*. New York: Wiley.
Nayfeh AH and Mook DT (1979) *Nonlinear Oscillations*. New York: Wiley.
Nayfeh SA, Nayfeh AH and Mook DT (1995) Nonlinear response of a taut string to longitudinal and transverse end excitation. *Journal of Vibration and Control* 1: 307–334.
O'Reilly O and Holmes PJ (1992) Nonlinear nonplanar, nonperiodic vibrations of a string. *Journal of Sound and Vibration* 153: 413–435.
Perez-Polo MF and Perez-Molina M (2007) A generalized mathematical model to analyze the nonlinear behaviour of a controlled gyroscope in gimbals. *Nonlinear Dynamics* 48: 129–152.
Rega G (2004) Nonlinear vibrations of suspended cables-part I: modelling and analysis and part-II: deterministic phenomena. *Applied Mechanics Reviews* 57: 443–478, 479–514.
Roberts JB (1985) Estimation of nonlinear ship roll damping from free decay data. *Journal of Ship Research* 29: 127–138.
Rubin MB and Gottlieb O (1996) Numerical solutions of forced vibration and whirling of a nonlinear string using the theory of a Cosserat point. *Journal of Sound and Vibration* 197: 85–101.
Sanders JA and Verhulst F (1985) *Averaging Methods in Nonlinear Dynamics*. New York: Springer.
Senesh M, Wolf A and Gottlieb O (2010) Nonlinear model-based estimation of rigid-body motion via an indirect measurement of an elastic appendage. *Journal of Vibration and Acoustics* 132(011007): 1–12.
Todd MD and Vohra ST (1998) Measurement and analysis of complex modulated motions in a weakly nonlinear system. *Physica D* 120: 329–345.
Tritton DJ (1977) *Physical Fluid Dynamics*. Oxford University Press: Oxford, UK.
Tritton DJ (1986) Ordered and chaotic motion of a forced spherical pendulum. *European Journal of Physics* 7: 162–169.
Tritton DJ and Groves M (1999) Lyapunov exponents for the Miles' spherical pendulum equations. *Physica D* 126: 83–98.

Appendix

Consider the following form for a quasilinear equation of motion:

$$x_{tt} + f_G(A,\ \Psi)x_t + f_H(A,\ \Psi)x = 0, \tag{A1}$$

where $f_G(A,\ \Psi)$ and $f_H(A,\ \Psi)$ define the system damping and stiffness backbone properties which are functions of the slowly varying instantaneous amplitude $A(\varepsilon t)$ and phase $\Psi(\varepsilon t)$. Substitution of an assumed harmonic solution form $x = A\sin(\Psi)$ where $\Psi = [(\bar{\omega} + \tilde{\omega}(\varepsilon t))t]$, into the equation of motion (A1) and application of a straightforward harmonic balance procedure (Nayfeh and Mook, 1979) yields:

$$\begin{aligned} &\sin(\Psi) : \mathrm{A}_{tt} - A\Psi_t^2 + f_G A_t + f_H A = 0 \\ &\cos(\Psi) : A\Psi_{tt} + 2A_t\psi_t + f_G A\Psi_t = 0. \end{aligned} \tag{A2}$$

Note that the harmonic balance equations (A2) enable isolation of the unknown amplitude damping and stiffness backbone properties and their determination as a function of the instantaneous amplitude, phase and their derivatives. We thus define the slowly varying phase derivative via the instantaneous frequency $\Psi_t = \omega = (2\pi F)$, and rearrange the second equation of the harmonic balance to yield the required damping backbone:

$$f_G = -(2A_t/A + F_t/F). \tag{A3}$$

We recall from linear vibration theory that the damping backbone property is equivalent to $f_G = (2\tilde{\zeta}\omega)$ where $\tilde{\zeta}$ is the damping ratio and ω is the radial frequency. Thus, the damping backbone can readily be put in its final functional form:

$$\tilde{\zeta} = -[A_t/A + F_t/(2F)]/(2\pi F). \tag{A4}$$

We note that this result is identical to that previously obtained via the Hilbert transform (Gottlieb and Feldman, 1997).

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